

Vertex Coverings by Coloured Induced Graphs— Frames and Umbrellas

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Abstract

A graph G homogeneously embeds in a graph H if for every vertex x of G and every vertex y of H there is an induced copy of G in H with x at y . The graph G uniformly embeds in H if for every vertex y of H there is an induced copy of G in H containing y . For positive integer k , let $f_k(G)$ (respectively, $g_k(G)$) be the minimum order of a graph H whose edges can be k -coloured such that for each colour, G homogeneously embeds (respectively, uniformly embeds) in the graph given by $V(H)$ and the edges of that colour. We investigate the values $f_2(G)$ and $g_2(G)$ for special classes of G , in particular when G is a star or balanced complete bipartite graph. Then we investigate $f_k(G)$ and $g_k(G)$ when $k \geq 3$ and G is a complete graph.

Keywords: homogeneously embeds, uniformly embeds, k -edge-colouring, bicliques, cliques, stars

1 Introduction

Motivated by results on the so-called framing number, Entringer et al. [3] determined the minimum order of a graph where every vertex belongs to both a clique of cardinality m and an independent set of cardinality n . They called this minimum $f(m, n)$.

Theorem 1 (Entringer et al. [3]) *For $m, n \geq 1$,*

$$f(m+1, n+1) = \left\lceil (\sqrt{m} + \sqrt{n})^2 \right\rceil = m + n + \left\lceil 2\sqrt{mn} \right\rceil.$$

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The result is a special case of results about framing number (see below) and was also generalised by Füredi, Mubayi and West [4]. Galvin [5] gave another proof of the result and observed that it could be viewed as a result about set systems.

In this paper we view Theorem 1 as a result about two-colouring the edges. In particular, $f(m, n)$ is the minimum order of a graph whose edges can be two-coloured with colours red and blue such that every vertex is in a clique of cardinality m whose edges are all red and every vertex is in a clique of cardinality n whose edges are all blue. We extend this concept by replacing clique with an arbitrary graph, and by replacing two colours with multiple colours.

We will need the concepts of homogeneous and uniform embedding introduced by Chartrand, Gavlas and Schultz [1]. They defined a graph G to be *homogeneously embedded* in a graph H if for every vertex x of G and every vertex y of H there is an induced copy of G in H with x at y . The graph G is said to be *uniformly embedded* in H if for every vertex y of H there is an induced copy of G in H containing y . Chartrand et al. [1] called a graph F of minimum order in which G can be homogeneously embedded a *frame* of G , and the order of F they called the *framing number* $fr(G)$ of G .

Let $k \geq 1$ and let G be a graph. We say that a graph H is $(G; k)$ -*homogenizable* if the edges of H can be coloured with k colours, say colours $1, \dots, k$, such that for all i , G homogeneously embeds in the graph given by $V(H)$ and the i^{th} coloured edges. A graph H of minimum order that is $(G; k)$ -homogenizable we call a $(G; k)$ -*frame*, and the order of H we call the $(G; k)$ -*framing number* of G , denoted by $f_k(G)$. When $k = 1$, $f_k(G)$ is the (original) framing number $fr(G)$ of G . Note that by duplicating a vertex and the edges incident with it, it follows that for every $n \geq f_k(G)$ there exists a graph H on n vertices that is $(G; k)$ -homogenizable.

Similarly, we say that a graph H is $(G; k)$ -*coverable* if the edges of H can be coloured with k colours, say colours $1, \dots, k$, such that for all i , G uniformly embeds in the graph given by $V(H)$ and the i^{th} coloured edges. A graph H of minimum order that is $(G; k)$ -coverable we call a $(G; k)$ -*umbrella*, and the order of H we call the $(G; k)$ -*umbrella number* of G , denoted by $g_k(G)$. When $k = 1$, the graph G itself is trivially a $(G; k)$ -umbrella, and so $g_1(G) = p(G)$, where $p(G)$ denotes the order of G . Again, for every $n \geq g_k(G)$ there exists a graph H on n vertices that is $(G; k)$ -coverable.

In Section 2 we observe some simple general results about frames and umbrellas. In Section 3 we determine $f_2(G)$ when G is a star and derive bounds on it when G is a biclique, while in Section 4 the corresponding results for g_2 are obtained. Finally, in Section 5, we investigate $f_k(G)$ and $g_k(G)$ when G is a complete graph.

For notation and graph theory terminology we in general follow [2]. Specifically, $\delta(G)$ and $\Delta(G)$ denote the minimum and maximum degrees of G . We will mostly consider two colours, and mostly these two colours will be red and white. If the edges of graph H are coloured with red and white, then the spanning subgraph of H containing all red (respectively, white) edges is denoted by R_H (respectively, W_H).

A preliminary version of this paper appeared as [6].

2 General Bounds

In this section we observe some simple bounds and some special cases.

The following simple lower bounds on $f_k(G)$ and $g_k(G)$ are useful for small graphs.

Lemma 2 *Let $k \geq 1$ be an integer and let G be a graph. Then*

- (a) $f_k(G) \geq k \cdot \Delta(G) + 1$ (with strict inequality if $\Delta(G)$ is odd and k is even), and
- (b) $g_k(G) \geq \Delta(G) + (k - 1)\delta(G) + 1$.

Proof. (a) Let H be a $(G; k)$ -homogenizable graph. Then every vertex of H has at least $\Delta(G)$ neighbours in each of the k colours, and so $|V(H)| \geq k\Delta(G) + 1$. If k is even this value is odd; so if $\Delta(G)$ is odd, equality in the bound is impossible.

(b) Let H be a $(G; k)$ -coverable graph. Since H contains an induced copy of G in the first colour, there is a vertex v of H with $\Delta(G)$ neighbours in the first colour. Now there is an induced copy of G in H containing v in each of the other colours, and so v has at least $\delta(G)$ neighbours in each of the other colours. Hence, $|V(H)| \geq \Delta(G) + (k - 1)\delta(G) + 1$. $\square$

As a simple example of these parameters, we calculate them for the path on 3 vertices.

Fact 3 $f_2(P_3) = 5$ and $g_2(P_3) = 4$.

Proof. By Lemma 2(a), $f_2(P_3) \geq 5$. On the other hand, K_5 is $(P_3; 2)$ -homogenizable, as one can two-colour the edges so that each colour induces a 5-cycle, and so $f_2(P_3) \leq 5$.

By Lemma 2(b), $g_2(P_3) \geq 4$. On the other hand, K_4 is $(P_3; 2)$ -coverable, as one can two-edge colour the edges so that each colour induces a path P_4 , and so $g_2(P_3) \leq 4$. $\square$

There are a few classes of graphs where the result is simple or related to other questions. For vertex-transitive graphs, the concepts homogenizable and coverable are the same. So if G is a vertex-transitive graph, then $f_k(G) = g_k(G)$. In particular, for the complete graph $f_k(K_n) = g_k(K_n)$, and as a special case of Theorem 1 we have:

Fact 4 *For $n \geq 2$, $f_2(K_n) = g_2(K_n) = 4(n - 1)$.*

If G is a self-complementary graph, then $g_2(G) = p(G)$. In particular, we have the following result.

Fact 5 $g_2(P_4) = 4$ and $g_2(C_5) = 5$.

This can be generalised when the graph is regular.

Lemma 6 *Let G be an r -regular graph on p vertices, and let nG denote the graph which is the union of n vertex-disjoint copies of G . If K_{np} is edge-decomposable into copies of nG , then $g_k(G) = np$ where $k = (np - 1)/r$.*

Proof. First we note that G has $pr/2$ edges, and that $\binom{np}{2}/(npr/2) = k$. Since nG packs into K_{np} , it thus follows that $g_k(G) \leq np$. On the other hand, By Lemma 2(b), $g_k(G) \geq r(np - 1)/r + 1 = np$. $\square$

This lemma implies that when the number of colours is large (relative to the size of the graph), the values are often given by results on designs. For example:

Theorem 7 *If q is a prime power, then $f_{q+1}(K_q) = q^2$.*

Proof. This follows from the existence of affine planes in this case [9]. $\square$

Finally in this section we consider general upper bounds for f_k and g_k . The first is a simple generalisation of a bound from [1], while the second uses the construction from [4] but with colours added.

Lemma 8 *For all graphs G of order $p > 1$ it holds that:*

- (a) $f_k(G) \leq kp(2p + 1)$, and
- (b) $g_k(G) \leq 2k(p - 1)$.

Proof. (a) Suppose we have colours $1, \dots, k$. Take $k(2p + 1)$ copies of G and label them $G_i^{(c)}$, for $1 \leq c \leq k$ and $1 \leq i \leq 2p + 1$. Then colour the edges of each $G_i^{(c)}$ with colour c . If v is a vertex of G , then the vertex of $G_i^{(c)}$ that corresponds to v we shall describe by the short-hand “the vertex $v \in G_i^{(c)}$ ”; similarly with sets of vertices.

Now we add edges. For colours $1 \leq c, d \leq k$, copies $1 \leq i \leq 2p + 1$ and vertex labels $1 \leq j, l \leq p$,

join $v_j \in G_i^{(c)}$ to $N(v_{j+l}) \subseteq G_{i+l}^{(d)}$ by edges coloured d ,

where $j + l$ is taken modulo p , and $i + l$ is taken modulo $2p + 1$. Thus the graph induced by $v_j \in G_i^{(c)}$, together with the vertices of $G_{i+l}^{(d)}$ except for v_{j+l} , is a copy of G coloured d —and v_j plays the role of v_{j+l} . Thus we have constructed a k -coloured graph on $kp(2p + 1)$ vertices in which G homogeneously embeds for each colour.

(b) Suppose we have colours $1, \dots, k$. Let x be any vertex of G and let Y be the neighbours of x . Take $2k$ disjoint copies of $G - x$. For each colour i , colour two copies of $G - x$ with colour i , call them A_i^0 and A_i^1 .

Now for each pair of colours $i < j$ and $e \in \{0, 1\}$, add edges as follows:

join all of A_i^e to $Y \subseteq A_j^e$ by colour j , and join all of A_j^e to $Y \subseteq A_i^{1-e}$ by colour i .

If $v \in A_i^e$, then $V(A_j^e) \cup \{v\}$ induces a copy of G coloured j . So every vertex appears in an induced copy of G with every colour, and thus $g_k(G) \leq 2k(p - 1)$. $\square$

3 Frames for Stars and Bicliques

As observed earlier, $f_1(G)$ is the framing number $fr(G)$ of G . Thus we have the following result from [1].

Theorem 9 (Chartrand et al. [1]) *For $b \geq 1$, $f_1(K_{1,b}) = 2b$.*

In fact, it is easy to see that $f_1(K_{a,b}) = 2b$ for all $a \leq b$. In this section, we determine $f_2(K_{a,b})$ for stars and some other complete bipartite graphs. We shall refer to a star $K_{1,b}$ simply as a b -star. We need the following lemmas; the first is a trivial extension of the above theorem.

Lemma 10 *If G is a graph such that every vertex is the centre of an induced b -star, then $p(G) \geq 2b$.*

Proof. Let v be any vertex and assume it is the centre of an induced b -star F_v . Let w be any end-vertex of F_v and let F_w be an induced b -star centered at w . Since $V(F_v) \cap V(F_w) \subseteq \{v, w\}$, the vertex w must be adjacent to at least $b - 1$ vertices that do not belong to F_v . Thus, $p(G) \geq 2b$. $\square$

The next two lemmas provide lower bounds on the 2-colour framing number of a star depending on whether the number of leaves is even or odd.

Lemma 11 *Let b be even. There is no graph of order $5b/2 - 1$ which can be two-coloured such that every vertex is the centre of an induced b -star in both colours.*

Proof. Suppose there exists such a graph, call it H . Let v be any vertex of H . Let A denote the end-vertices of an induced red b -star centered at v , and let B denote the end-vertices of an induced white b -star centered at v .

Now, consider a vertex $a \in A$ and an induced red b -star centered at a . This star includes at least $b/2 + 1$ vertices from B , as it cannot include another vertex from A , and so a is adjacent in R_H to at least $b/2 + 1$ vertices from B . Similarly, every vertex in B is adjacent in W_H to at least $b/2 + 1$ vertices in A , but there are not enough edges between A and B , a contradiction. $\square$

Lemma 12 *Let $b \geq 5$ be odd. There is no graph of order $(5b + 1)/2$ which can be two-coloured such that every vertex is the centre of an induced b -star in both colours.*

Proof. Suppose there exists such a graph, call it H . Let v be any vertex of H . Let A denote the end-vertices of an induced red b -star centered at v , and let B denote the end-vertices of an induced white b -star centered at v . Let $X = V(H) - A - B - \{v\}$. Then $|X| = (b - 1)/2$.

Claim 1. *Both R_H and W_H are b -regular and thus triangle-free.*

PROOF OF CLAIM 1. We prove that R_H is b -regular. The proof for W_H is identical.

Suppose that v has degree more than b in R_H . Then there is a vertex x_1 of X that is adjacent to v in R_H . Now, consider a vertex $a \in A$ and an induced red b -star centered at a . This star includes at least $(b+1)/2$ vertices from B , as it cannot include another vertex from A , nor can it use both v and x_1 . Similarly, for a vertex $b \in B$, any induced white b -star centered at b includes at least $(b-1)/2$ vertices from A .

This implies that every vertex of A is adjacent in R_H to exactly $(b+1)/2$ vertices from B , and every vertex of B is adjacent in W_H to exactly $(b-1)/2$ vertices from A . Further, the edges from B to X are all white, and the edges from A to $X - \{x_1\}$ are all red.

Next, consider an induced red b -star centered at x_1 . This star uses no vertex of B . If it uses a vertex of $\{v\} \cup (X - \{x_1\})$, then it can use no vertex of A , a contradiction. Hence, x_1 must be adjacent in R_H to every vertex of A .

Finally, consider an induced white b -star centered at $a \in A$. This star contains only vertices from $A \cup B$. Thus, a is the centre of an induced white $(b+1)/2$ -star in the graph induced by A . This is true for every vertex of A . Hence, by Lemma 10, $|A| \geq b+1$, a contradiction. $\square$

Claim 2. *X is an independent set in both R_H and W_H .*

PROOF OF CLAIM 2. Suppose there is a red edge between two vertices x_1 and x_2 of X . Then, as in the above proof, an induced red b -star centered at a vertex $a \in A$ includes at least $(b+1)/2$ vertices from B , as it cannot use both x_1 and x_2 . Similarly, an induced white b -star centered at a vertex $b \in B$ includes at least $(b-1)/2$ vertices from A . So, every vertex of A is adjacent in R_H to exactly $(b+1)/2$ vertices from B , and every vertex of B is adjacent in W_H to exactly $(b-1)/2$ vertices from A . Furthermore, the edges from B to X are all white, and the edges from A to $X - \{x_1, x_2\}$ are all red.

Next, consider induced red b -stars centered at x_1 and x_2 . Neither of these stars uses any vertex of B . If one of these stars uses a vertex of $\{v\} \cup (X - \{x_1, x_2\})$, then it can use no vertex of A , and is too small, a contradiction. Thus, x_1 and x_2 must be adjacent in R_H to at least $b-1$ vertices of A . Thus, since $b \geq 3$, there is a vertex of A that is adjacent to both x_1 and x_2 in R_H , which contradicts Claim 1. $\square$

Now, by Claim 2, X is an independent set of H . By Claim 1, v is adjacent to no vertex of X in H , and so $X \cup \{v\}$ is an independent set of H . Since each vertex of X has degree (at least) $2b$ in H , all edges between the set X and the set $A \cup B$ are present in H . Since v was arbitrary, every vertex of H is in an independent set of cardinality $(b+1)/2$ which is joined by edges to all other vertices. It follows that these independent sets partition the vertices of H .

Thus, $p(H) = (5b + 1)/2$ must be a multiple of $(b + 1)/2$. However,

$$\frac{(5b + 1)/2}{(b + 1)/2} = 5 - \frac{4}{b + 1},$$

and so $4/(b + 1)$ is an integer, which is impossible since $b \geq 5$. $\square$

We note that if $b = 3$ then the above proof leads to the extremal example of the cube. We are now in a position to determine $f_2(G)$ when G is a star.

Theorem 13

$$f_2(K_{1,b}) = \begin{cases} 5b/2 & \text{for } b \text{ even,} \\ (5b + 1)/2 & \text{for } b = 3, \\ (5b + 3)/2 & \text{for } b \neq 3 \text{ odd.} \end{cases}$$

Proof. The result is true for $b = 1, 2$ by Facts 3 and 4.

Assume b even, $b \geq 4$. The lower bound follows from Lemma 11. To construct a $(K_{1,b}; 2)$ -frame, consider the expansion of K_5 given by the lexicographic product $K_5(\bar{K}_{b/2})$. That is, take K_5 and replace each vertex by $b/2$ mutually nonadjacent copies of it. Now, K_5 can be two-coloured so that each colour induces a C_5 ; extend this colouring to the expansion of K_5 . This shows that the expansion of K_5 , which has order $5b/2$, is $(K_{1,b}; 2)$ -homogenizable, and so $f_2(K_{1,b}) \leq 5b/2$.

Assume b odd, $b \geq 5$. The lower bound follows from Lemma 12. To construct a $(K_{1,b}; 2)$ -frame, take the graph constructed above for $b + 1$ and delete one vertex. The result is a $(K_{1,b}; 2)$ -homogenizable graph of order $(5b + 3)/2$, and so $f_2(K_{1,b}) \leq (5b + 3)/2$.

Finally, consider $b = 3$. Lemma 2(a) shows that $f_2(K_{1,3}) \geq 8$. To construct a $(K_{1,3}; 2)$ -frame, note that $K_{1,3}$ homogeneously embeds in the cube Q_3 . Further K_8 contains two edge-disjoint copies of the cube Q_3 , and so $f_2(K_{1,3}) \leq 8$. (If the first cube has disjoint 4-cycles $u_1u_2u_3u_4u_1$ and $v_1v_2v_3v_4v_1$ as well as edges u_iv_i , then the other cube has disjoint 4-cycles $u_1v_2u_3v_4u_1$ and $v_1u_2v_3u_4v_1$ as well as edges u_iv_{i+2} , with arithmetic modulo 4.) $\square$

A consequence of the above result and its proof is the framing number of lop-sided bicliques:

Corollary 14 For $2 \leq a \leq b/2$,

$$f_2(K_{a,b}) = \begin{cases} 5b/2 & \text{for } b \text{ even,} \\ (5b + 3)/2 & \text{for } b \text{ odd.} \end{cases}$$

We turn next to balanced bicliques. We start with a special case.

Theorem 15 $f_2(C_4) = 7$.

Proof. By Lemma 2(a), $f_2(C_4) \geq 5$. If a graph F has 5 vertices and C_4 homogeneously embeds in F , then F has at least 6 edges. (Consider any 4-cycle: the remaining vertex has degree at least 2.) So two such F cannot fit into 5 vertices.

Suppose a graph H on 6 vertices is two-coloured such that C_4 homogeneously embeds in each colour. Then one colour has at most 7 edges, say red. Consider any induced red 4-cycle, say $v_1v_2v_3v_4v_1$. The remaining two vertices, say v_5 and v_6 , each have degree at least 2 in R_H , and so in fact must have degree exactly 2 and must be adjacent. Their other neighbours in R_H must also be adjacent and so we may assume they are v_1 and v_2 . But even if all the other edges are white, there cannot be a white 4-cycle containing v_1 , a contradiction. It follows that $f_2(C_4) \geq 7$.

For the upper bound, consider the graph X formed by taking a 6-cycle, adding a new vertex v and making it adjacent to three independent vertices on the cycle. (The graph X is also obtained by deleting a vertex from the cube Q_3 .) Then two copies of X fit into K_7 (since two copies of Q_3 fit into K_8 —see case $b = 3$ in the proof of Theorem 13). $\square$

An open question is what happens for general $K_{b,b}$. By expanding the construction for C_4 above, it follows that $f_2(K_{b,b}) \leq 7b/2 + O(1)$. It is not hard to show a lower bound of $3b$ but we believe that equality holds in the upper bound.

Lemma 16 $f_2(K_{b,b}) \geq 3b$.

Proof. Consider a graph H with its edges two-coloured with red and white. Suppose H contains an induced red $K_{b,b}$ and an induced white $K_{b,b}$, say R and W . Then we claim that one partite set from one of the bicliques is disjoint from the other biclique. For suppose W intersects both partite sets of R ; then the vertices of $R \cap W$ are in the same partite set of W and so the other partite set of W is disjoint from R . $\square$

With considerably more effort, we can show that $f_2(K_{b,b}) \geq 3.1b$.

Conjecture 1 For b even, $f_2(K_{b,b}) = 7b/2$.

4 Umbrellas for Stars and Bicliques

In this section we consider the behaviour of g_2 for lop-sided bicliques. We give exact results when the biclique is sufficiently lop-sided.

Theorem 17 Assume $b \geq a$. Then $g_2(K_{a,b}) \geq \min(2b, b + 3a)$ with equality if $b \geq 2a$.

Proof. We establish the lower bound first. Let G be a graph which is two-coloured with red and white such that each vertex is in a $K_{a,b}$ of each colour. Suppose G has fewer than $2b$ vertices.

Let A_R denote the set of vertices which are on the a -side of some red $K_{a,b}$, and B_R the set of vertices which are on the b -side of some red $K_{a,b}$. Since a vertex of A_R is adjacent to at least b vertices in R_G , and a vertex of B_R is nonadjacent to at least $b - 1$ vertices in R_G , and G has less than $2b$ vertices, a vertex cannot be in both A_R and B_R . So, A_R and B_R partition $V(G)$. Further, a red edge joining two vertices in A_R , or joining two vertices in B_R , would not be in a red $K_{a,b}$, and so we may discard it. It follows that we may assume A_R and B_R are both independent sets in R_G .

Similarly, define A_W and B_W for the other colour white, and it follows that A_W and B_W partition $V(G)$. Now, since a vertex in A_R is incident with b red edges, and a vertex in A_W is incident with b white edges, by the constraint on the order of G it follows that A_W and A_R are disjoint. Let $X = B_W \cap B_R$ and let H denote the (bipartite) graph induced by $A_R \cup A_W$.

Now, every vertex v in A_W needs to be part of a red $K_{a,b}$. Let L_v denote the vertices of A_W in this biclique, and M_v the vertices of A_R . Similarly, for a vertex v in A_R , let L_v denote the vertices of A_R in its white $K_{a,b}$, and M_v the vertices of A_W . In particular, $|M_v| \geq a$ —every vertex of A_W has at least a red neighbours in A_R and every vertex in A_R has at least a white neighbours in A_W . Since there are at least $a|A_W|$ red edges and at least $a|A_R|$ white edges, but at most $|A_W||A_R|$ edges in total, a bit of calculus shows that $|A_W| + |A_R| \geq 4a$.

Now, let ℓ denote the minimum of $|L_v|$ over all v , say attained for vertex $v \in A_W$. First note that $|X| \geq b - \ell$. On the other hand, a vertex of M_v has a white neighbours in A_W , which are necessarily disjoint from L_v , and so both partite sets of H have at least $\ell + a$ vertices. Thus $p(H) \geq 2a + 2\ell$.

Hence $p(H) \geq \max(4a, 2a + 2\ell)$ and so $p(G) \geq \max(4a + b - \ell, 2a + b + \ell) \geq 3a + b$. That is, we have shown that $g_2(K_{a,b}) \geq \min(2b, b + 3a)$.

If $3a \leq b$, then an umbrella is given by the following construction. Take the wheel $C_4 + K_1$. On the 4-cycle, colour the edges alternately with red and white. For the edges incident with the dominating vertex, colour red the edges to one pair of opposite vertices on the 4-cycle and colour white the other pair. Then, expand each vertex on the 4-cycle to be an independent set of a vertices, and expand the dominating vertex to be an independent set of $b - a$ vertices. The resulting graph has $3a + b$ vertices and every vertex is in both a red and a white $K_{a,b}$.

If $2a \leq b \leq 3a$, then an umbrella is given by the following construction. Take K_4 with vertex set $\{v_1, v_2, v_3, v_4\}$. Then colour the three edges v_1v_2 , v_2v_3 and v_3v_4 red, and the other three edges white. Then expand both vertices v_1 and v_4 to independent sets of $\lfloor b/2 \rfloor$ vertices, and expand both vertices v_2 and v_3 to independent sets of $\lceil b/2 \rceil$ vertices. The resulting graph has $2b$ vertices and every vertex is in both a red and a white $K_{\lfloor b/2 \rfloor, b}$. $\square$

As a consequence of the above (and Fact 4), we have the following formula for the star:

Fact 18 $g_2(K_{1,b}) = b + 3$ unless $b = 2$ when $g_2(K_{1,b}) = b + 2$.

For balanced bicliques, the value of g_2 is of course the same as f_2 . In particular:

Fact 19 $g_2(C_4) = 7$.

5 Cliques

Since the complete graph is vertex-transitive, $f_k(K_n) = g_k(K_n)$. The problem for two colours has already been solved (Fact 4). So we focus on three colours.

In some instances, the question of $f_k(K_n)$ is connected to packings and design theory via Lemma 6. This was exploited in Theorem 7. As another example, for $k \equiv 1 \pmod{3}$, since Kirkman triple systems of order $2k + 1$ exist [8], it holds that $f_k(K_3) = 2k + 1$.

Theorem 20 $f_3(K_3) = g_3(K_3) = 9$.

Proof. From the comment about Kirkman triple systems above or Theorem 7, it follows that $f_3(K_3) \leq f_4(K_3) = 9$. We need to show that $f_3(K_3) > 8$.

Suppose there exists a 3-colouring of the edges of a graph H of order 8 such that every vertex is in a triangle of each colour. Assume the colour red is used on the fewest edges: then at most 9 edges are red. Further, every vertex is incident with at least two edges of each colour. Thus, the graph H_R induced by the red edges has 8 vertices, at most 9 edges, maximum degree at most 3 and every vertex is in a triangle. But this is readily seen to be impossible. $\square$

For a 4-clique, the best upper bound we have is that $f_3(K_4) = g_3(K_4) \leq 16$, the value for five colours given in Theorem 7.

In general, we have only the upper bound from Lemma 8:

$$f_3(K_m) = g_3(K_m) \leq 6(m - 1).$$

Lower bounds seem difficult. Recent progress is to be found in [7].

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